



The bioinspired vision sensors developed by Dr Chai's team can adapt to varying brightness with an effective range of up to 199dB. The human retina can adapt to environments under sunlight to starlight, with a range of about 160dB (Credit: PolyU)

They are designed to look like the rod and cone cells in the human eye, which are responsible for detecting dim and strong light, respectively. As a result, the sensors can identify things in a variety of lighting conditions, as well as flip between and adapt to varied brightness levels—with a range that is even greater than the human eye!

3D printing can now be done in seconds

EPFL researchers have shown a light-based 3D-printing technology that is around 30 times faster than traditional additive manufacturing, producing items in 20 seconds that would ordinarily take more than 10 minutes. Unlike traditional 3D printing, which involves placing blobs of material on a substrate and allowing them to solidify, EPFL's light based technology involves pouring the resin into a contain-



The team printed this little Yoda figure in about 20 seconds (Credit: EPFL)

er and spinning it. Then, light is shined on the container at different angles, causing the resin to solidify wherever the accumulated energy in the resin exceeds a given level. It's a very precise method and can produce objects at the same resolution as existing 3D-printing techniques.

World's first SNN developed at IIT Bombay

A team from IIT Bombay has collaborated to create the world's first spiking neuro-synaptic (SNN) core on 45nm SoI (silicon on insulator) technology. The Liquid State Machine, a spoken word recognition system, was used in this development. It is based on the premise that the neural



(Left to Right) Prof. Maryam Shojaei Baghini, Vivek Saraswat, Prof. Udayan Ganguly, Ajay Singh holding the test board with the neural network chip (Credit: Ankit Bende)

network produces a neuronal reservoir similar to a liquid reservoir. This method was made by including numerous hardware peculiarities in 'ideal' software implementations to demonstrate the hardware implementation's viability. The circuit implementation included a 20-neuron input layer that sent the neural spike signal into a 36-neuron reservoir with 256 synapses connecting neurons in a random recurrent fashion.

Microprocessors can now be powered by photosynthesis

Researchers from the University of Cambridge's Department of Biochemistry used algae to power an Arm CPU for six months! The scientists created an AA-battery-sized gadget that houses an algae called *Synechocystis* that naturally harvests energy from the sun through photosynthesis. This biological photovoltaic cell produces power continually, even in low light or at night, because the algae it contains appears to continue processing energy after the sun has set. The device was made of common, sturdy,



The renewable biological photovoltaic cell (Credit: Paolo Bombelli)